Yilin Wang

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Yilin Wang | Dylan-Wyl10 | 0000-0002-6002-0535
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Research Interests

Seeking **Research Scientist / Applied Scientist / ML Engineer** roles in autonomous driving and intelligent transportation systems:

- **Robotaxi & AV fleet operations** – routing, dispatching, and repositioning at network scale, with fleets doubling as mobile sensing platforms; via MILP and learning-based combinatorial optimization.
- **LLM/VLM agents & world models for AV & embodied AI** – trajectory-planning foundation models for behavior simulation (sim agents), world-model construction, and safety-critical scenario generation & closed-loop validation.
- **Agentic system implementation & deployment** – multi-agent architectures, tool orchestration (MCP), and agent evaluation for production workflows, from smart-city & robotaxi operations to automated research pipelines.
- **Cooperative perception & V2X systems** – from algorithms to real-world field deployment.

Education

- **Purdue University** *Aug.2021 - Spring 2027 (Expected)*
  Doctor of Philosophy West Lafayette, US
  - Research Area: Connected and Automated Vehicle, Cooperative Perception, Physics-informed Machine Learning, Network Operation and Control
  - GPA: 3.8/4.00
- **Carnegie Mellon University** *Aug.2019 - Dec.2020*
  Master of Science Pittsburgh, US
  - Area: Smart City, Deep Learning, Natural Language Processing
  - Course: Deep Learning, Reinforcement Learning, Geo-Informatics System
  - GPA: 3.87/4.00
- **Tongji University** *Aug.2015 - Jun.2019*
  Bachelor of Science Shanghai, China
  - Major: Civil Engineering (Area of Geo-technical and Underground Engineering)

Working Experience

- **Inchor(SaferDrive AI)** *Dec.2025 - Jun.2026*
  Simulation Development Engineer Intern Ann Arbor, MI
  - Develop and maintain TeraSim, an open-source city-scale traffic simulation platform for autonomous vehicle safety validation.
  - Implement a full-stack Driving Foundation Model into product.
  - Lead joint project with NVIDIA, and City of San Jose on Smart City Operation with agent
  - Initial and develop Agent-Sim, an agentic World Model for autonomous driving safety validation.

Patents and Publications

J=Journal, C=Conference

[J.1] **Yilin Wang**, Yiheng Feng* (2026). **RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring**. Under review at *Transportation Research Part C*.

[J.2] Hanlin Chen, Vamsi K Bandaru, **Yilin Wang**, Mario A Romero, Andrew Tarko, Yiheng Feng* (2024). Cooperative Perception System for Aiding Connected and Automated Vehicle Navigation and Improving Safety in *Transportation Research Record*, 2678(12), 1498-1510. https://doi.org/10.1177/03611981241252779

[J.3] **Y. Wang** and Y. Feng* (2024). IDM-Follower: A Model-Informed Deep Learning Method for Car-Following Trajectory Prediction in *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 6, pp. 5014-5020, June 2024, doi: 10.1109/TIV.2024.3367654

[J.4] Ya-Dong Xue, Wei Zhang, **Yi-Lin Wang***, et al. (2023). Serviceability evaluation of highway tunnels based on data mining and machine learning: A case study of continental United States in *Tunneling and Underground Space Technology*, Volume 142, 2023, 105418, ISSN 0886-7798, https://doi.org/10.1016/j.tust.2023.105418

[C.1] **Yilin Wang**, Yiheng Feng* (2026). **RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring**. Scheduled for presentation at the *INFORMS 2026 Annual Meeting*.

[C.2] **Yilin Wang**, Yiheng Feng* (2026). **A Cooperative Perception Based Dynamic Vehicle Routing Framework for Urban Traffic Monitoring**. In *Transportation Research Board Annual Meeting 2026*, Accepted by Poster Presentation.

[C.3] **Yilin Wang**, Yiheng Feng* (2024). Dynamic Routing of Connected and Automated Vehicles for Improving Network Coverage. In *Transportation Research Board Annual Meeting 2024*, Accepted by Poster Presentation.

PROJECTS

- **TeraSim: City-Scale World Model for Safety Testing on AV and Physical AI** *Dec.2025 - Present*
  Core Developer | Intership | SaferDrive AI(Inchor)
  - **Overview**: A city-scale simulation platform serving as the foundational infrastructure for agentic world models, enabling naturalistic and adversarial testing of autonomous vehicles at urban scale.
  - **Core Module – NNDE (Neural-network Naturalistic Driving Environment)**: Behavioral-layer foundation module of TeraSim that integrates a transformer-based motion model to drive city-scale, naturalistic AV behavior generation.
  - **Core Module – Agentic Test Engineer**: A multi-agent framework powered by LLMs and VLMs that performs autonomous scenario mutation to safety-critical corner cases for AV testing.
  - **Milestone**: 1. 2026-03: Active deployment with **NVIDIA** and **City of San Jose** as an *Agentic Incident-to-Operation* pipeline, bringing TeraSim from simulation into real-world city operations. 2. 2026-06: First draft release on Agent-Sim, utilizing LLMs and VLMs for safety critical senario generation.
  - **Tech Stack**: Python, SUMO, PyTorch, CUDA, Docker,

- **Developing A Cooperative Perception System** *Jan.2023 - Present*
  Project Leader | Funded by Center of Connected and Automated Transportation
  - **Phase 1**: A Cooperative Perception System for CAV Navigation and Improving Safety
    - **Objective**: Implement a cooperative perception system to improve road safety.
    - **Contribution**:
      1. We build a cooperative perception system connected by V2X system between roadside LiDAR and CAV sensors.
      2. We develop a detecting and navigating algorithm on CAV with by sensor data fusion pipeline.
      3. We build and conduct a field testing demonstrating the safety benefits in detecting occluded VRUs through cooperative perception system.
    - **Conclusion**: Cooperative perception system helps CAVs to plan safer (i.e., higher post-encroachment time) and smoother (i.e., lower deceleration rates) trajectories.
    - **Outcome**:
      1. Publication: **Cooperative Perception System for Aiding Connected and Automated Vehicle Navigation and Improving Safety**
  - **Phase 2**: An Infrastructure-vehicle Cooperative Perception System
    - **Objective**: Develop a cooperative perception system including multiple vehicles, infrastructure sensors, and drones.
    - **Contribution**: Extract and processing multi-sensor data (Lidar, Camera, etc.)to vehicle trajectories. Release open-source dataset for futural research.
    - **Methodology**: Multi-sensor sensor fusion from infrastructure sensor, drone-view camera, and vehicle sensor.

- **Cooperative Perception based Operation for CAVs to Improve Traffic Data Collection** *Apr.2023 - Jun.2026*
  Individual Research Project [GitHub]
  - **Project 1**: Dynamic Routing on Connected and Automated Vehicles for Improving Network Coverage
    Apr.2023 - Aug.2024
    - **Research Question**: How to model the CAV network coverage and operate routes to improve data collection.
    - **Contribution**:
      1. We formulate the CAV routing problem considering the network coverage as one objective.
      2. We propose a heuristic algorithms with greedy search to solve the multi-objective optimization.
      3. We conduct a simulation experiment in a grid network to verify the feasibility and relations between travel cost and network coverage.
    - **Methodology**: heuristic optimization, SUMO simulation, link-based travel time estimation.
    - **Outcome**: Accepted by Transportation Research Board (TRBAM) 2024 as poster presentation.
  - **Project 2**: A Cooperative Perception based Dynamic Vehicle Routing Framework For Urban Traffic Monitoring
    Sept.2024 - Jun.2026
    - **Research Question**:
      1. A higher granularity of road network representation is needed.
      2. A proper performance metric is needed to evaluate traffic monitoring impact.
      3. There is a dynamic effect between CAV routes decision and partial network observations.
    - **Contribution**:
      1. We apply Cell Transmission Model (CTM) for traffic state prediction and road network representation considering CAV detection maneuver.
      2. We propose a comprehensive MILP formulation considering the performance of traffic monitoring as well as dynamic vehicle routing in one explict equation.

3. We conduct a microscopic simulation and evaluations under various scenarios.

* **Outcome**: RoboSense is under review at *Transportation Research Part C*; scheduled for presentation at the *INFORMS 2026 Annual Meeting*.

- **IDM-Follower: A Physics-Informed Neural Network Model for Trajectory Prediction** *Jan.2022 - Feb.2024*
  Individual Research Project [GitHub]
  - **Research Question**: How to integrate explicit car-following model into learning-based model to improve training and performance.
  - **Contribution**:
    1. We introduce a physics-informed neural network (PINN) model that integrates intelligent driving model (IDM).
    2. The proposed model exhibits robustness against real-time GPS noise.
  - **Methodology**: The loss function combined by physical laws and ground-truth difference is utilized to train a customized attention-based VAE model.
  - **Outcome**:
    1. Presentation: Research Presentation on IEEE-IV 2024 at Jeju Island, KR.
    2. Publication: **IDM-Follower: A Model-Informed Deep Learning Method for Car-Following Trajectory Prediction**

- **Developing V2X System for Traffic Signal Control** *Sept.2021 - Jun.2026*
  Multiple Projects, Funded by US-DOT, Michigan DOT, and Leidos Inc.
  - **Part 1: Smart Intersection Project** *Sept.2021 - Jun.2026*
    * **Overview**: Implement CV2X system for Transit Signal Priority (TSP) in City of Ann Arbor.
    * **Contribution**:
      1. Redevelop Multi-Modal Intelligent Traffic Signal System (MMITSS). [GitHub]
      2. Maintain and update versions on road-side and vehicle-side applications.
      3. Field testing, installation, and data processing for buses and real intersections. [GitHub]
  - **Part 2: Michigan DOT ATCMTD Project** *Feb.2023 - present*
    * **Overview**: Implementing MMITSS system in Q-Line in Detroit, MI.
    * **Contribution**: Update MMITSS to accommodate NTCIP 1211 operation protocol. [GitHub]
  - **Part 3: CDA-Sim Update: Integrating MMITSS on a Co-simulation Platform** *Dec.2024 - Jun.2025*
    * **Overview**: Integrating MMITSS into CARMA-CARLA-SUMO co-simulation.
    * **Contribution**:
      1. New features on time synchronization, message brokers, and parallel processing to support MMITSS in CDA-Sim platform. [GitHub]
      2. Build integration testing and demo with released configuration.

Skills

- **Programming Languages:** Python, C++, MATLAB, JavaScript
- **Data Science & Machine Learning:** PyTorch, CUDA, Hugging Face
- **Cloud Technologies:** AWS, SSH
- **DevOps & Version Control:** Linux, Docker, Conda
- **Cooperation Platform & Project Management:** Jira, Slack, GitHub
- **Other Tools & Technologies:** Microsoft Office, Auto-CAD, ArcGIS
- **Research Skills:** Critical Thinking, Research Talk, Comprehensive Writing, Literature Review

Honors and Awards

- **Honorable Mention** *2018*
  Mathematical Contest in Modeling of USA [Web]

Additional Information

Languages: English (Proficiency), Chinese Mandarin (Native)
Interests: Soccer, Skateboarding, Range and Gaming

REFERENCES

1. **Yiheng Feng**
   Assistant Professor, Civil and Construction Engineering
   Assistant Director, Center for Road Safety (CRS)
   Lyles School of Civil and Construction Engineering, Purdue University
   Relationship: Ph.D Advisor

2. **Pingbo Tang**
   Associate Professor, Civil and Environment Engineering
   Carnegie Mellon University
   Relationship: Master Research Advisor

Contact details for references available upon request.